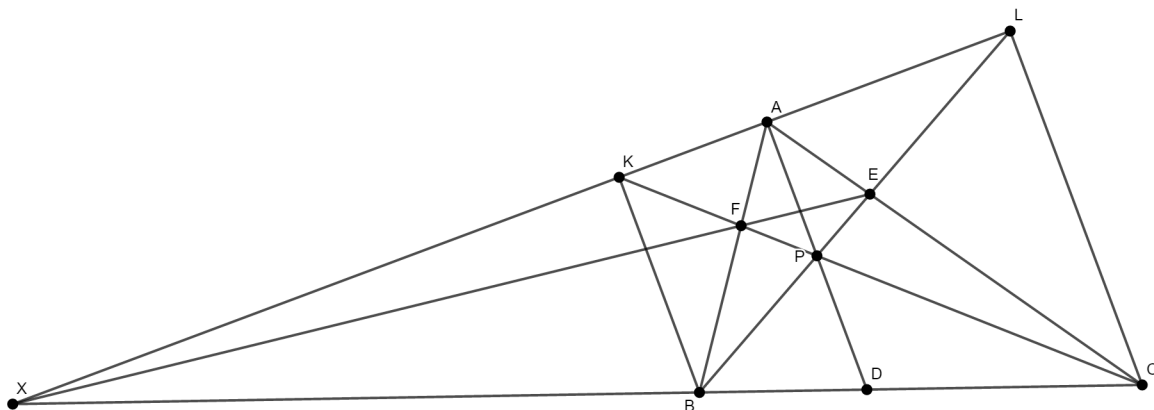


4.3.16



Let CK and BL meet at P , and AP meet BC at D .

Obviously, $\triangle ABK \sim \triangle ACL$ and $BK \parallel CL$. Then $\frac{AK}{AL} = \frac{BK}{CL} = \frac{BP}{PL}$, so $AP \parallel CL$, APD is the angle bisector of $\angle BAC$.

By angle bisector theorem and Ceva's theorem,

$$\frac{BX}{XC} \times \frac{CE}{EA} \times \frac{AF}{FB} = \frac{BA}{AC} \times \frac{CE}{EA} \times \frac{AF}{FB} = \frac{BD}{DC} \times \frac{CE}{EA} \times \frac{AF}{FB} = 1$$

By Menelaus theorem, E, F, X are collinear.